Long Nguyen

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Summary

I am a Software Engineer at Nokia working on AI-RAN, focusing on the L1 aspect of 5G mobile communications. My daily work revolves around real-time digital signal processing, CUDA-accelerated high-performance computing, and modern C++ engineering for mission-critical telecommunications systems.

Experience

Software Engineer Jan 2026 – Now
Nokia - Full time *Espoo, Finland*

- Developing high-performance L1 physical layer components for Nokia's next-generation AI-RAN platform.
- Implementing high-parallel computing kernels in CUDA and C++ for real-time 5G signal processing.
- Optimising latency and throughput constraints in mobile communication systems.
- Skills: CUDA, C++20, 5G NR L1, Parallel Computing, AI-RAN

Trainee July 2025 – December 2025
Nokia - Full time *Espoo, Finland*

- Developed visualization tools to support 5G communication system modeling and performance analysis.
- Skills: CI development, Scrum, Software Development skill

Master's Thesis worker Jan 2025 – Jun 2025
Nokia - Full time *Espoo, Finland*

- Topic: 5G Physical Layer Channel Conditions Classification Using Machine Learning Techniques
- Responsible for building a deep learning model for 5G NR TDL/CDL channel condition classification
- Skills: Tensorflow, Communication Engineering

Bachelor's Thesis worker Dec 2023 – Jan 2024
Tampere University - Apprenticeship *Tampere, Finland*

- Topic: Transforming Frontal to Lateral: Chest X-ray View Conversion Using CycleGAN
- Responsible for building a CycleGAN model with UNet - ResNet architecture with/without Self-Organized Operational Layers
- Skills: TensorFlow, CycleGAN

Teaching Assistant Sep 2023 – May 2024
Tampere University - Contract *Tampere, Finland*

- Course: COMP.CS.140. Programming 3: Interfaces and Techniques (5 cr)
- Skills: Java, API, Time management
- Responsible for Coditoriums, Supporting students, Evaluating students' exercises and projects

Research trainee Jan 2023 – May 2023
Tampere University - Apprenticeship *Tampere, Finland*

- Project: Automatization of image correction through the shaping of light
- Supervisors: Prof. Robert Flicker, Postdoctoral Researcher Rafael Barros
- Skills: Matlab, Genetic Algorithms, Photonics
- Responsible for applying Zernike polynomial and 4F system to a spatial light modulator (SLM)
- Apply genetic algorithms for modifying the imperfect output of the SLM.

Education

Aalto University Espoo, Finland
Master of Science (Technology) *Aug 2024 - Dec 2026*

- Cumulative GPA: 4.5/5 (Currently, 75 ECTS)
- Specialization: Machine Learning, Data Science and Artificial Intelligence
  - Focus on Machine Learning theory, Deep Learning, Reinforcement Learning, and High-performance computing

Tampere University Tampere, Finland
Master's Programme in Computing Sciences *Aug 2024 - July 2026*

- Specialization: Signal Processing and Machine Learning
- GPA: 4.43/5 (123 ECTS)
- Major: Signal Processing and Machine Learning
  - Advanced knowledge of Machine Learning, Image, Signal, and Audio Processing
- Minor: Communications and Networking
  - Base knowledge of Communication, Computer Networking, Software Design

Tampere University Tampere, Finland
Bachelor of Science and Engineering - Computing and Electrical Engineering track *Aug 2022 – Jun 2024*

- GPA: 4.51/5 (180 ECTS)
- Major: Signal Processing and Machine Learning
  - Base knowledge of Machine learning, Image and Audio Processing, Embedded Systems, Communication Engineering, and Control Systems
- Minor: Software Systems
  - Base knowledge of Cyber Security, Human Interaction Technology, and Software Development
- Free choice studies:
  - Base knowledge of Git, and Research Training

Skills

Programming Languages	:	Python, C, C++, Java, Scala, Matlab, R
Frameworks	:	React.js, Node.js, Mocha, Spark, TensorFlow
Databases	:	MongoDB, SQLite
Dev Tools	:	Apache, Qt, Xilinx SDK, Git, SonarQube
Time Management	:	244+ ECTS/2 years - Normal Duration 120 ECTS/2 years

Projects

Quantitative RL Trading System Aalto University | Master's Thesis
textit 2025 – Present

- Master's thesis research framework developed at Aalto University for building, training, replaying, and inspecting reinforcement-learning trading policies (PPO / SAC). Features timestamped market-data pipelines, causal multi-timeframe feature calculations, broker-neutral interfaces, simulated execution, and rigorous evaluation with full experiment management.
- Skills: Python, Reinforcement Learning, PPO / SAC, Market Microstructure, FastAPI, Docker

5G NR Physical Layer Channel Classification Nokia | Master's Thesis
textit Jan 2025 – Jun 2025

- Open research framework for 5G NR physical layer channel analysis and classification (TDL / CDL channel condition models). Investigated physically grounded LOS/NLOS detection under severe domain shift in wireless propagation environments to support adaptive L1 baseband processing.
- Skills: Python, TensorFlow, 5G NR L1, Wireless AI

Solibri Autorun Custom View Tool €3 000 Scholarship Award
textit Jan 2024 – May 2024

- Engineered a desktop application with Java Spring Boot and JavaFX designed to eliminate syntax and semantic errors in complex XML automated building information modeling (BIM) model checking. Featured dynamic directory scanning, automated input constraint validation, Figma UI design, and SonarQube code quality assurance.
- Skills: Java, Spring Boot, JavaFX, XML, SonarQube, Agile Scrum
- Scholarship for the project: 3000€

Transforming Frontal to Lateral: Chest X-ray View Conversion Bachelor's Thesis
textit Dec 2023 – Jan 2024

- Synthesised lateral chest radiographs from frontal views using a custom CycleGAN framework with UNet–ResNet architecture and Self-Organised Operational Layers (SOOL). Overcame cross-projection ambiguities in medical imaging to enhance anatomical and diagnostic fidelity.
- Skills: TensorFlow, CycleGAN, Medical Imaging, ResNet, Computer Vision

Certifications & Achievements

- 100% Tuition Fee Scholarship: Tampere and Aalto University
- €3 000 Project Scholarship: Solibri Autorun Custom View Tool
- MATLAB Fundamentals Certification

Additional

Languages : English (Professional), Vietnamese (Bilingual), Finnish (Elementary)